

Optimization-based machine learning for computational imaging

Plug-and-play (PnP)

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Inference framework: feed-forward NN

$$\begin{aligned}(\forall \mathbf{x}^{[0]} \in \mathbb{R}^{N_0}) \quad \mathbf{x}^{[K]} &= \mathfrak{L}_{\Theta}^K(\mathbf{x}^{[0]}) \\ &= \mathfrak{T}_{\Theta_K} \circ \dots \circ \mathfrak{T}_{\Theta_1}(\mathbf{x}^{[0]}),\end{aligned}$$

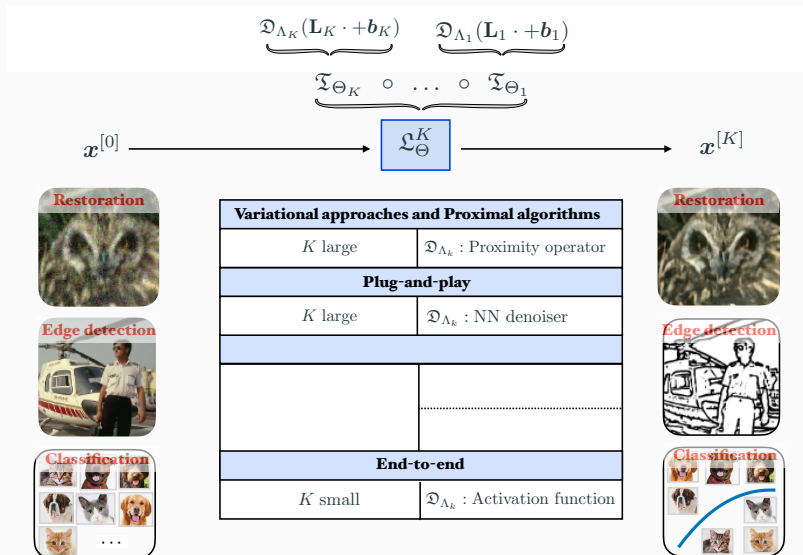
Layer/iteration

$$\mathfrak{T}_{\Theta_k} : \mathbb{R}^{N_{k-1}} \rightarrow \mathbb{R}^{N_k} : \mathbf{x} \mapsto \mathfrak{D}_{\Lambda_k}(\mathbf{L}_k \cdot + \mathbf{b}_k),$$

- $\mathbf{L}_k : \mathbb{R}^{N_{k-1}} \rightarrow \mathbb{R}^{N_k}$: linear operator,
- $\mathbf{b}_k \in \mathbb{R}^{N_k}$: shift parameter,
- $\mathfrak{D}_{\Lambda_k} : \mathbb{R}^{N_k} \rightarrow \mathbb{R}^{N_k}$: nonlinear operator parametrized by Λ_k .

Parameters: $\Theta = \cup_{k=1}^K \Theta_k$ with $\Theta_k = \{\Lambda_k, \mathbf{L}_k, \mathbf{b}_k\}$.

Unified framework



Deep learning – General framework

- Database : $\mathcal{S} = \{(\bar{x}_i, y_i) \in \mathbb{R}^N \times \mathbb{R}^M \mid i \in \mathbb{I} \cup \mathbb{J} \text{ such that } y_i = \bar{x}_i + \varepsilon\}$
- Prediction function : $d_{\Theta}(y_i) = \eta_K(L_K \dots \eta_1(L_1 y_i + b_1) \dots + b_K)$

Variational formulation versus Plug and play

- **Forward-Backward**: $x^{[k+1]} = \text{prox}_{\tau\theta g \circ D} (x^{[k]} - \tau A^{\top} (Ax^{[k]} - z))$
- **FB-PnP**: $x^{[k+1]} = d_{\Theta} (x^{[k]} - \tau A^{\top} (Ax^{[k]} - z))$

Remark

- Principle that can be applied to other algorithmic schemes PnP-ADMM, PnP-DR, HQS, ...

FB-PnP:

$$\mathbf{x}^{[k+1]} = \mathbf{d}_{\Theta} \left(\mathbf{x}^{[k]} - \tau \mathbf{A}^{\top} (\mathbf{A} \mathbf{x}^{[k]} - \mathbf{z}) \right)$$

Build a denoiser: MMSE denoiser $\mathbf{d}_{\Theta}(\mathbf{z}) = \mathbb{E}\{\mathbf{x} | \mathbf{z} = \mathbf{x} + \sigma\epsilon\}$

- Training :

$$\min_{\theta} \mathbb{E}_{\substack{\sigma \sim \mathcal{U}([0, \sigma_{max}]) \\ \mathbf{x} \sim p \\ \epsilon \sim \mathcal{N}(0, \text{Id})}} \{ \|\mathbf{d}_{\Theta}(\mathbf{x} + \sigma\epsilon) - \mathbf{x}\|^2 \}$$

- Tweedie Formula :

$$\mathbf{d}_{\Theta} = \text{Id} + \sigma^2 \nabla \log p$$

- **FB-PnP:**

$$x^{[k+1]} = d_{\Theta} \left(x^{[k]} - \tau A^{\top} (Ax^{[k]} - z) \right)$$

Build a denoiser: MAP denoiser

- Defining the estimate $d_{\Theta}(z)$ of $\bar{x}$ from z as the maximum of $p(x|z)$:

$$d_{\Theta}(z) \in \underset{x}{\operatorname{Argmin}} -\log p(z|x) - \log p(x).$$

For white Gaussian noise, $-\log p(z|x) = \frac{1}{2\sigma^2} \|z - x\|_2^2$.

Assume that $-\log p(x) = g(x)$. Then,

$$d_{\Theta}(z) = \operatorname{prox}_{\sigma^2 g}(z).$$

Remark

- g often unknown. This can explain the desire of replacing proximity operator by

- **FB-PnP:**

$$x^{[k+1]} = d_{\Theta} \left(x^{[k]} - \tau A^{\top} (Ax^{[k]} - z) \right)$$

Build a denoiser

- If $d_{\theta} = \text{prox}_{\tau\theta g}$ with g **convex** and $\tau < 2/\|A\|^2$, then convergence to $\hat{x} \in \underset{x}{\text{Argmin}} \frac{1}{2}\|Ax - z\|_2^2 + \theta g(x)$.

- **FB-PnP:**

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- If $d_{\theta} = \text{prox}_{\tau\theta g}$ with g **convex** and $\tau < 2/\|A\|^2$, then convergence to $\hat{x} \in \underset{x}{\text{Argmin}} \frac{1}{2}\|Ax - z\|_2^2 + \theta g(x)$.
- If $d_{\theta} = \text{prox}_{\tau\theta g}$ with g **nonconvex** and $\tau < 1/\|A\|^2$, then convergence to $\hat{x} \in \underset{x}{\text{Argmin}} \frac{1}{2}\|Ax - z\|_2^2 + \theta g(x)$. [Bolte, Sabach, Teboulle, 2014]

- **FB-PnP:**

$$x^{[k+1]} = d_{\Theta} \left(x^{[k]} - \tau A^{\top} (Ax^{[k]} - z) \right)$$

Build a denoiser

- d_{Θ} **built to be firmly non-expansive** by regularizing the training loss.

For instance $d_{\Theta} = \frac{\text{Id} + Q_{\Theta}}{2}$ with Q_{Θ} a non-expansive (1-Lipschitz) operator, then convergence to an inclusion problem when $\tau \leq 2/\|A\|^2$,

$$0 \in A^*(A\hat{x} - z) + \tau^{-1}(2(\text{Id} + Q_{\Theta})^{-1} - \text{Id})\hat{x}$$

[Pesquet, Repetti, Terris, Wiaux, 2021].

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[Pesquet, Repetti, Terris, Wiaux, 2021].

- **Training stage:** Find

$$\hat{\Theta} \in \underset{\Theta}{\text{Argmin}} \frac{1}{|\mathbb{I}|} \sum_{j \in \mathbb{I}} \ell(\bar{x}_j, d_{\Theta}(y_j)) + \lambda \max\{\|\nabla Q_{\Theta}(\tilde{x}_j)\|^2, 1 - \epsilon\}.$$

with $\lambda > 0$, $\tilde{x}_j = \rho_j \bar{x}_j + (1 - \rho_j)d_{\Theta}(y_j)$ and $\rho_j \sim \mathcal{U}([0, 1])$.

- **FB-PnP:**

$$x^{[k+1]} = d_{\Theta} (x^{[k]} - \tau A^{\top} (Ax^{[k]} - z))$$

Build a denoiser

- d_{Θ} built to be the proximal operator by regularizing the training loss.

If $d_{\Theta} = \text{Id} - \nabla \tilde{g}_{\theta}$ with $\nabla \tilde{g}_{\theta}$ β_g -Lipschitz with $\beta_g < 1$, there exists a (M -weakly convex with $M = \beta_g/(\beta_g + 1)$) potential g_{θ} such that $d_{\Theta} = \text{prox}_{g_{\theta}}$, then convergence to a minimization problem for $\tau < 2/(L_f + M)$.

[Hurault, Leclaire, Papadakis, 2022]

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with $\lambda > 0$.

- **FB-PnP:**

$$x^{[k+1]} = d_{\Theta} \left(x^{[k]} - \tau A^{\top} (Ax^{[k]} - z) \right)$$

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[Hurault, Leclaire, Papadakis, 2022]

- d_{Θ} is **built from an unfolded strategy** to compute $\text{prox}_{\|L \cdot\|_1 + \iota_C}$.

[Le, Foare, Pustelnik, 2022].

2- Avoid new training with an equivariant strategy

Equivariant network

Equivariant network

- d_θ is **invariant** to a transformation D if the output remains unchanged:

$$d_\theta(Dz) = d_\theta(z)$$

- d_θ is **equivariant** to a transformation D if the output changes in a corresponding way:

$$d_\theta(Dz) = D' d_\theta(z)$$

Particular case: If G acts on the same way in the output and input spaces, $d_\theta(Dz) = D d_\theta(z)$

Make d_θ $\mathcal{G}$ -equivariant

- Averaging over a group $\mathcal{G}$ of unitary matrix $\{D_g\}_{g \in \mathcal{G}}$:

$$d_{\theta, \mathcal{G}}(z) = \frac{1}{|\mathcal{G}|} \sum_{g \in \mathcal{G}} D_g^{-1} d_\theta(D_g z) \quad \rightarrow \quad \text{can be computationally demanding}$$

Equivariant Plug-and-play

- **FB-PnP:**

$$\mathbf{x}^{[k+1]} = \text{d}_{\Theta} \left(\mathbf{x}^{[k]} - \gamma \mathbf{A}^{\top} (\mathbf{A} \mathbf{x}^{[k]} - \mathbf{z}) \right)$$

- **Equivariant FB-PnP** [Terris, Moreau, Pustelnik, Tachella, 2024]

Sample $g_k \in \mathcal{G}$

$$\mathbf{x}^{[k+1]} = \mathbf{D}_{g_k}^{-1} \text{d}_{\Theta} \left(\mathbf{D}_{g_k} \mathbf{x}^{[k]} - \gamma \mathbf{A}^{\top} (\mathbf{A} \mathbf{x}^{[k]} - \mathbf{z}) \right)$$

Equivariant Plug-and-play

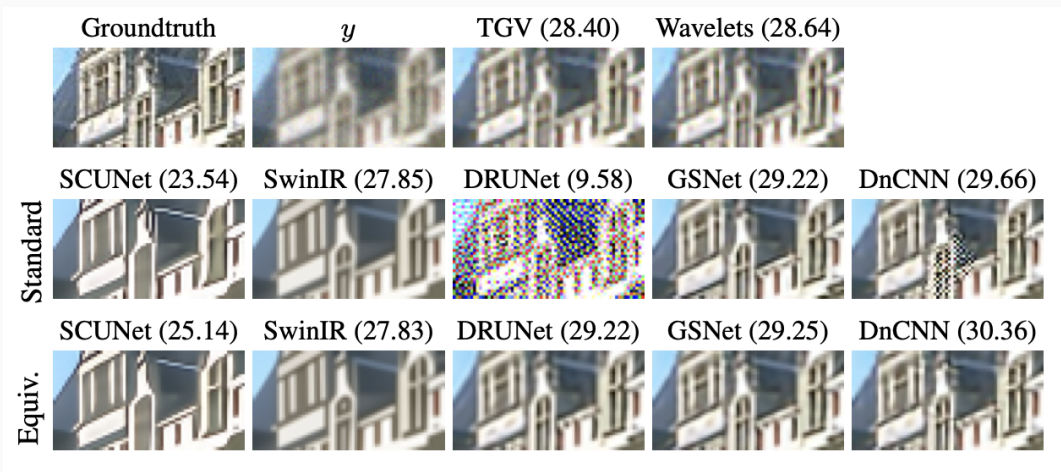
Proposition

$d_{\Theta}z = Wz$ be a linear denoiser with singular value decomposition $W = \sum_{i=1}^n \lambda_i u_i v_i^{\top}$ and $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n \geq 0$. If the principal component $u_1 v_1^{\top}$ is not $\mathcal{G}$ -equivariant, the averaged denoiser $d_{\Theta, \mathcal{G}}$ has a strictly smaller Lipschitz constant than d_{Θ} .

	Lipschitz DnCNN	DnCNN	SCUNet	SwinIR	DRUNet ($\sigma_d=0.01$)
Standard d_{Θ}	1.06	1.44	5.78	6.28	1.57
Equivariant $d_{\Theta, \mathcal{G}}$	0.92	1.18	4.19	4.05	1.26

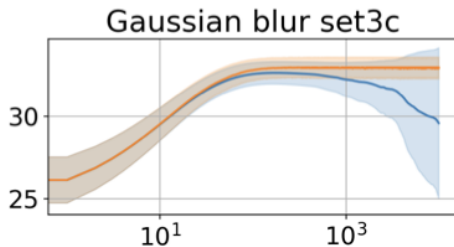
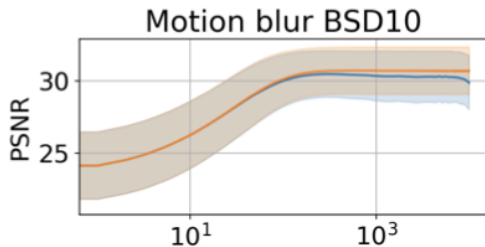
Lipschitz constant (i.e., the spectral norm of the Jacobian) of various denoisers averaged over 10 different patches of 64×64 pixels. Equivariant denoisers are obtained by averaging over the group of 90 degree rotations and reflections.

Numerical experiments Gaussian deblurring

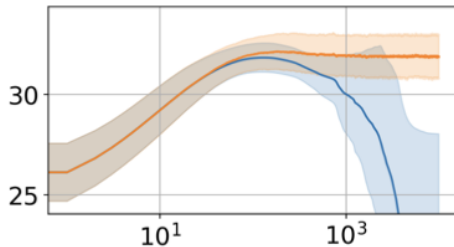
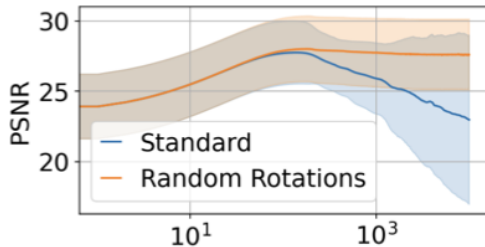


Numerical experiments deblurring

DnCNN



DRUNet



Conclusion

